FIG. 3 Exact-File Write Scope Isolation & Bounded Planner Handoffs

(a) Structured Plan & Bounded Worker Handoffs

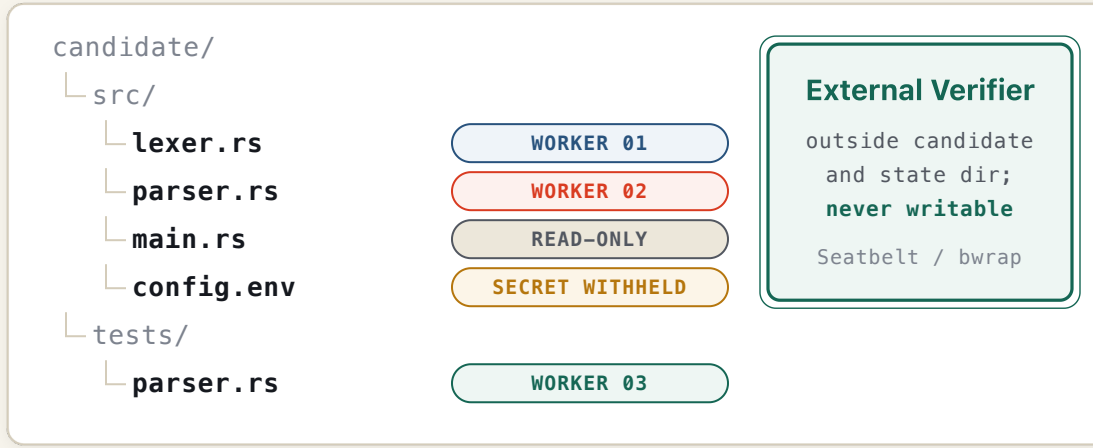
planner_response.json (≤ 64 KiB)

```
{"tasks": [
  {"task": "implement lexer",
   "write_paths": ["src/lexer.rs"],
   "context": "Keep UTF-8 offsets; <=8192 B"},
  {"task": "parse expressions",
   "write_paths": ["src/parser.rs"]},
  {"task": "cover parser invariants",
   "write_paths": ["tests/parser.rs"]}]}
```

(b) Whole-Plan Pre-Admission Invariants

Nonempty exact-file scope	Every coding task claims explicit files; no dirs
Canonical relative ASCII	Rejects .. traversal, //, symlinks, non-ASCII
Globally disjoint claims	No duplicate paths or ancestor/descendant overlap
Case-insensitive guard	Rejects ASCII case aliases (SRC/ vs src/)
Autonomy subset rule	All claims ⊆ autonomy.write_paths when set
Bounded handoff text	Optional context ≤ 8192 B/task; untrusted data

(c) Candidate Ownership & Secret Isolation Map



(d) Fail-Closed Claims (Abort Before Any Worker Starts)

`src/`	Directory is ancestor of src/parser.rs	ANCESTOR
`SRC/Parser.rs`	ASCII case alias of src/parser.rs	CASE ALIAS
`src/../.env`	Parent traversal + excluded secret	TRAVERSAL
`src//lexer.rs`	Non-canonical repeated path separator	SYNTAX
`context: 9 KiB`	Exceeds 8192-byte per-task handoff bound	HANDOFF CAP
`src/lexer.rs ×2`	Duplicate claim across coding tasks	DISJOINT

Source: src/runtime.rs (plan validation & worker dispatch), src/workspace.rs (SHA-256 mutation mutex & secret_like guard).